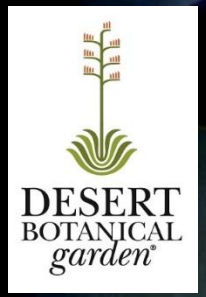
Becoming a Budding Botanist:

Partnering to better understand the flora of Arizona





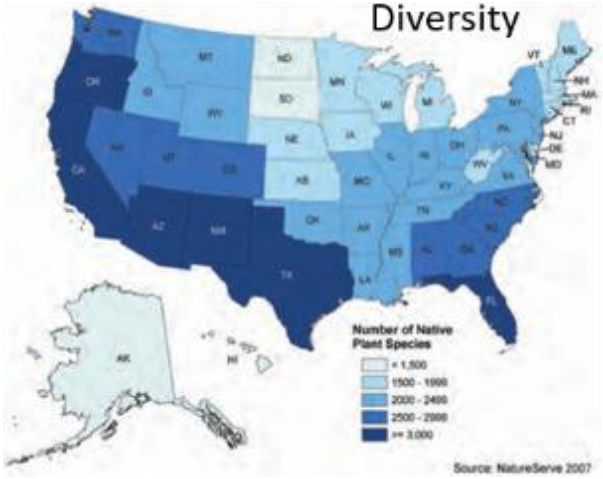
PAPAZ (Our) Goals

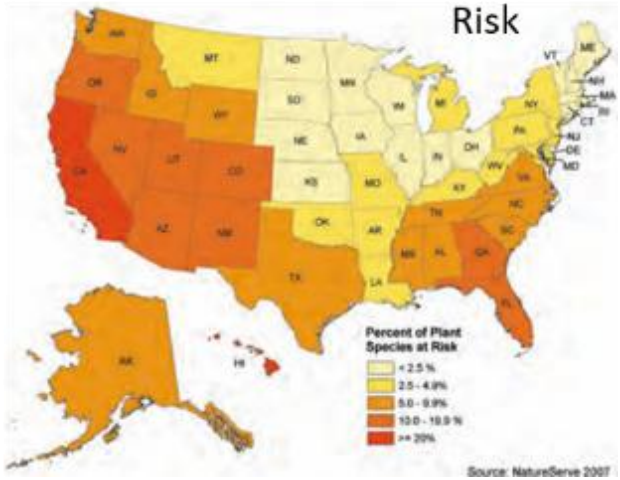




- To improve our scientific knowledge of AZ flora for land management, scientific, conservation, and educational purposes.
- To train, engage and educate members of the public as plant stewards.
- To provide a databased plant atlas located on the Southwest Environmental Information Network (SEINET)

Did you know?



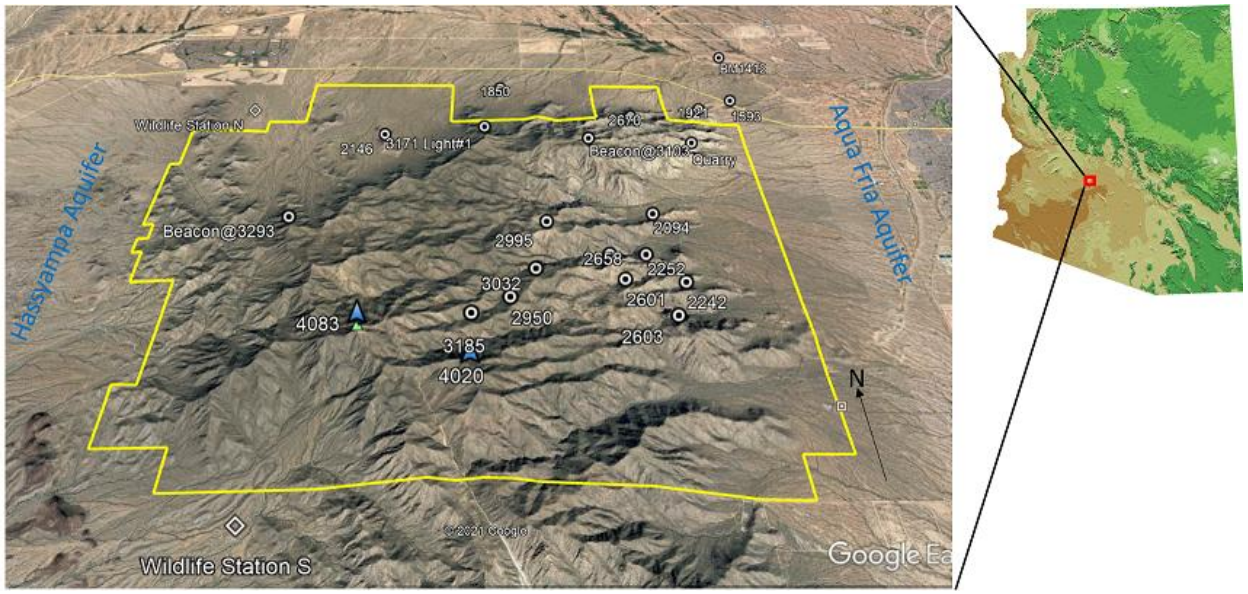


- Arizona is one of the top four states for floristic diversity
- Over the last 60 years an average of ca. 10 new species records have been reported annually

Key Unanswered Questions

- What areas of Arizona have the greatest floristic diversity?
- Are there new plant species yet to be discovered in Arizona?
- Where should we be conserving land?
- How can we provide decision-makers with sound, scientifically based information?

What the heck is a “flora”?



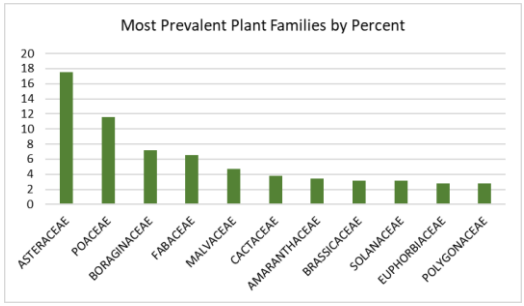
- A “flora”, or floristic inventory, is the scientific documentation of the plant life in a particular area.
- A flora includes field work, collecting specimens, documenting the specimens, plant identification, preserving and mounting the plants, and herbarium accessioning.

Why are floras important?

Amsonia kearneyana

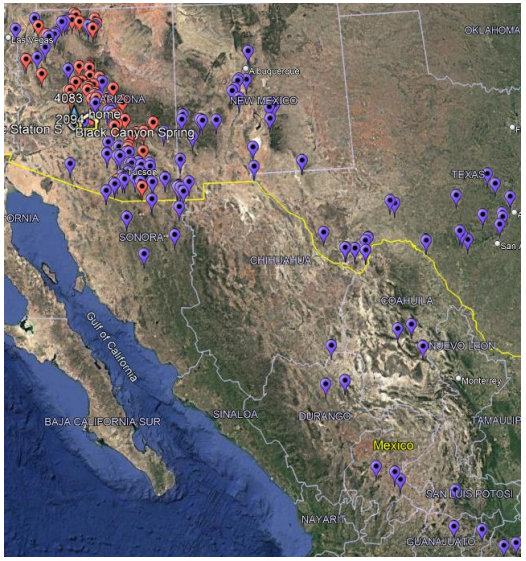


Identify significant populations in an area



Understanding flora composition of a region

Red vs Blue *Maurandella antirrhiniflora*



How populations are distributed

Carex atrospicata



Discover new species

- Floras provide a critical baseline understanding of the diversity of species in a place at a given time. We can't manage what we don't know!
- Floras allow scientists and land managers to assess change in biodiversity over time due to change (human-caused or natural).
- Floras are used by researchers to further our understanding of the plant world. Specimens are essentially read and studied.

What is my job?

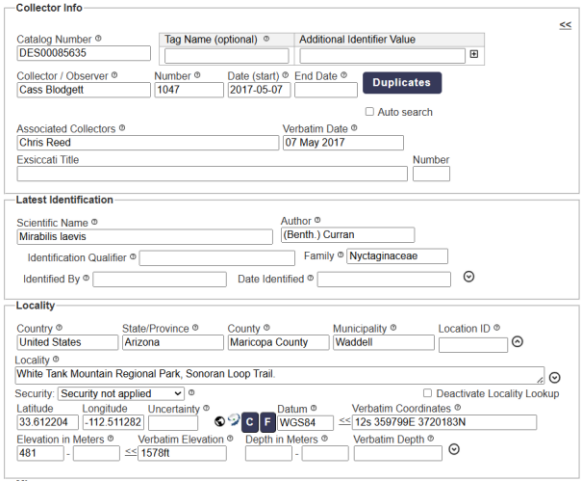


- Fieldwork (collecting data, voucher specimens, and lots of hiking!)
- With extra training (herbarium-based work at DBG)

Plant Identification



Data Entry



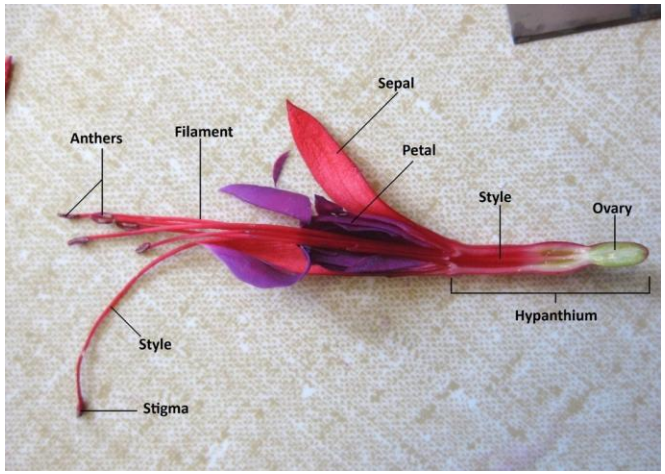
Specimen Mounting



Let's Get Started!
Basic Botany and Taxonomy

-or-

How you will never look at a plant in the same way!

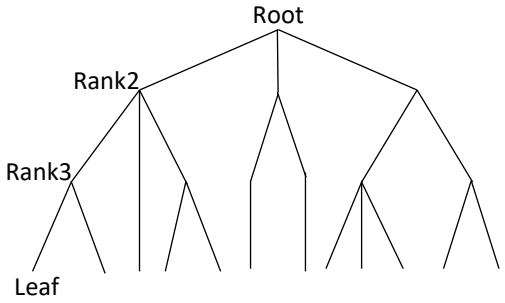




Understanding and naming our natural world is the first step to protecting and conserving the world's biodiversity. — Kirstin Phillips, MNA Botany Collections Manager

What is Taxonomy?

- The science of classifying, describing and naming organisms into hierarchical groups based on shared characteristics
  - Characters may be morphological, ecological, biochemical and genetic



A Tree Structure

Kingdom	Plantae	Plants	
Subkingdom	Viridiplantae	Green plants	
Division	Tracheophyta	Vascular plants	(vs. Bryophyta)
Subdivision	Spermatophytina	Seed plants	(vs. Polypodiophytina)
Class	Magnoliopsida	Flowering plants	(vs. Pinopsida)
Superorder	Magnolionae	Dicotyledons	(vs. Lillianae)
Order	Caryophyllales	Betalain pigments	
Family	Cactaceae	Cactus family	
Genus	Cylindropuntia	Cholla genus	
Species	Cylindropuntia fulgida (Engelm.) Knuth	Chainfruit cholla	



*ITIS

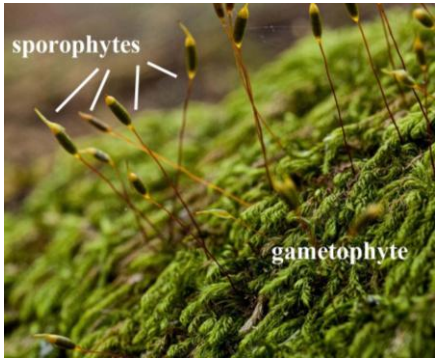
Alternation of Generations

Two life phases of plants. These phases vary in size and dependence on each other in different plant divisions

Gametophyte – sexual phase of life: generate sperm and eggs
fertilization takes place

Sporophyte – asexual phase of life: disperse propagules

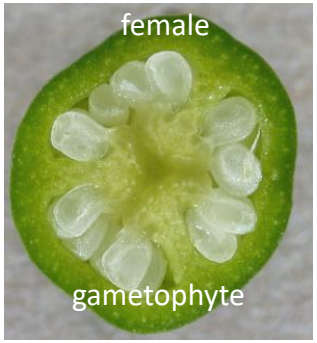
Moss



Fern



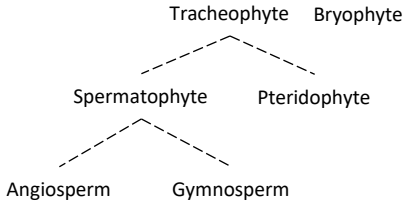
Flowering Plant







Major Plant Categories



- **Bryophytes** (moss-plant) – Zygote/embryo retention. Rizoids. Non vascular. Propagules are spores

  Moss, Liverwort

- **Pteridophytes** (feather-plant) – Vascular (xylem, phloem, lignin). Roots. Propagules are spores

  Ferns, Spike Moss

- **Spermatophytes** (Seed Plants) – Vascular, propagules are seeds with endosperm. Pollen/pollen tube – reduce water dependence for fertilization

  - **Gymnosperms** (Naked seeds) – Ovules exposed. Wind pollinated.

    Conifer, Ephedra

  - **Angiosperms** (Vesseled seeds) – Ovules enclosed in carpel. Insect pollinated (and wind).

    Flowering plants



480mya



400mya



350mya



135mya

What is a Species

- It is the lowest major taxonomic rank
  - Some species can be further categorized into subspecies and varieties
- *Theoretically,* a "species" is a series of populations that will interbreed and produce fertile seed – but there are many exceptions!

- Every species has a unique botanical name assigned to it



Phacelia crenulata
Scorpion-weed



Phacelia distans
Scorpion-weed



Amsinckia menziesii
Scorpion-weed

The Binomial System

- The binomial is composed of the genus and the specific epithet
- *Combined,* they form the species name, followed by the author name(s) to be precise
  - Genus must be a noun
  - Specific epithet must be an adjective or possessive

Eriastrum eremicum (Jepson) Mason

Genus | Specific epithet | Orig Author | New Author

Species



Plant Lifecycles

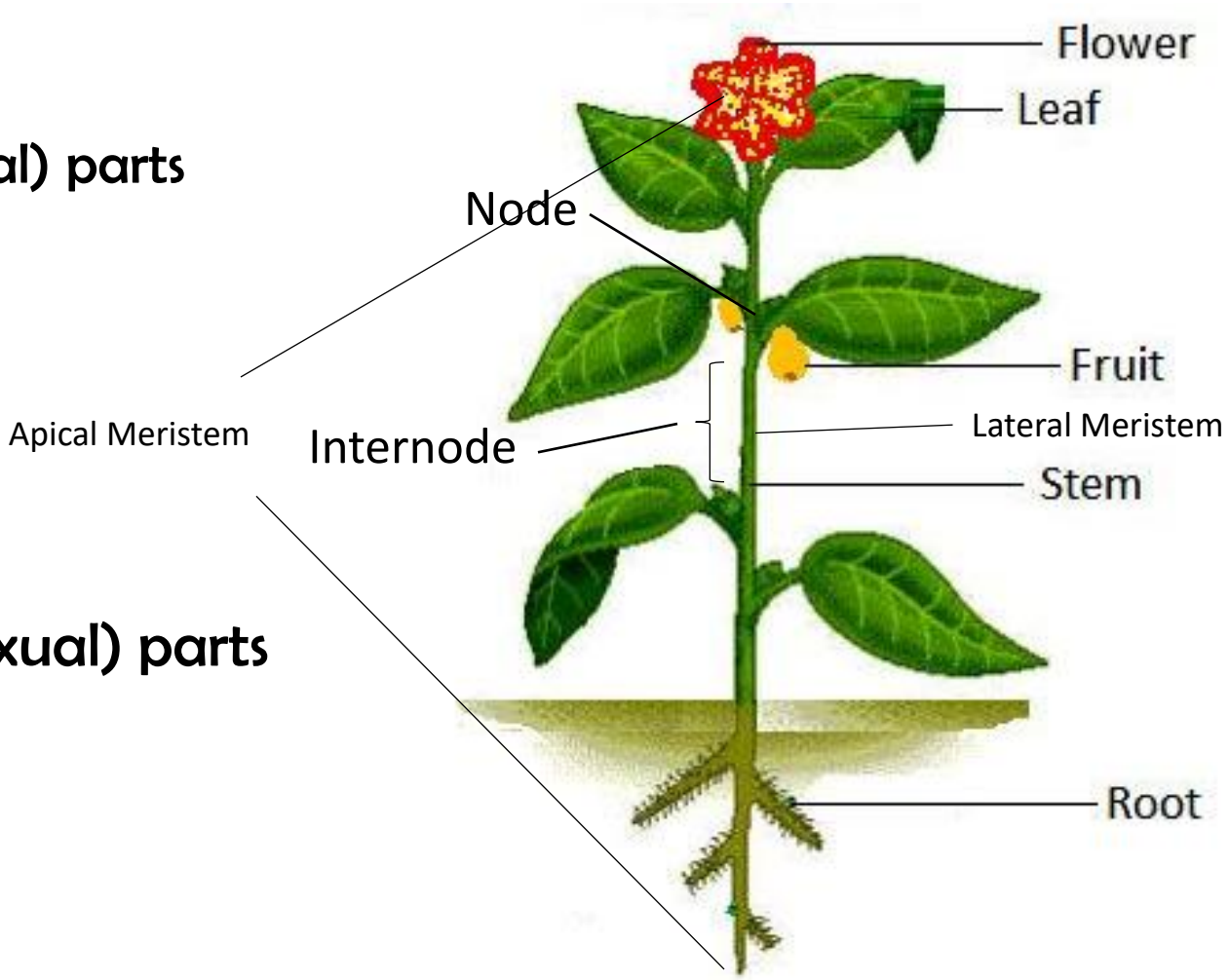
- Perennial - Plant with life cycle of multiple years.
  - Plants having woody structure - *phanerophyte*
  - Or lack woo but have a large nutrient storage root - *geophyte*
- Annual - Plant with life cycle less than one year - flowers once
  - Plants are usually small and herbaceous - fast growing.
- Monocarp - Plant that flowers just once and dies
  - But may live many years - Agave sp.
  - Or be short lived



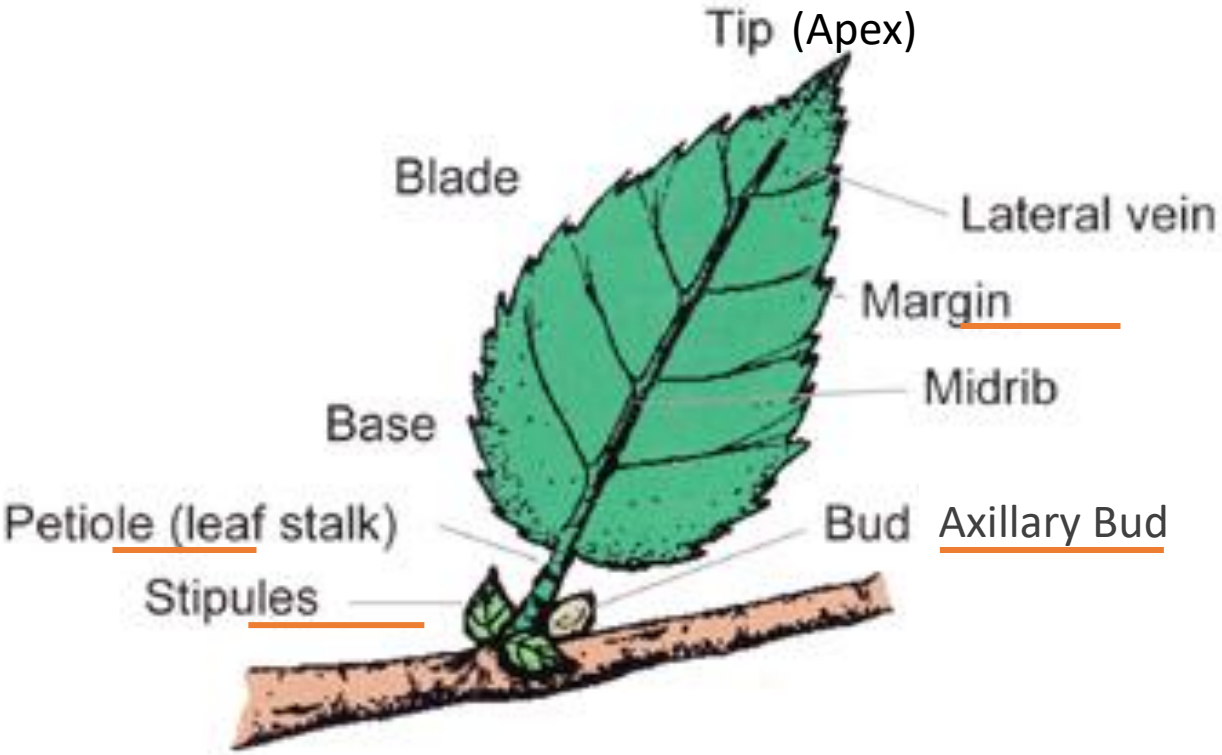


Plant Structure

- Vegetative (asexual) parts
  - Roots
  - Stems
    - Node
    - Internode
  - Leaves
- Reproductive (sexual) parts
  - Flowers
  - Fruits



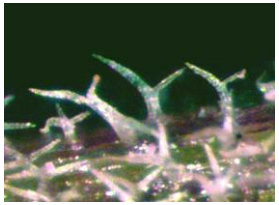
Leaf Structure



Leaf Hair



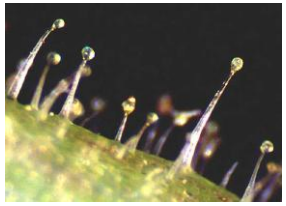
Simple



Branched



Stellate



Glandular

*Hair may be present on reproductive parts too

Leaf Shapes



Elliptic



Ovate



Obovate



Oblong



Lanceolate



Oblanceolate



Cordate



Deltoid



Reniform



Linear



Subulate



Aricular

Leaf Arrangement



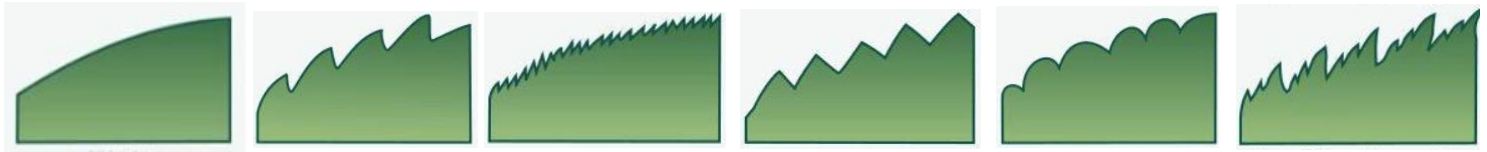
Alternate | Opposite | Opposite Decussate | Alternate Spiral | Whorled | Basal/Rosette

Leaf Attachment



Petiolate | Sessile | Sheathing | Clasping | Perfoliate

Leaf Margin



Entire | Serrate | Serrulate | Dentate | Crenulate | Doubly Serrate

Leaf Divisions

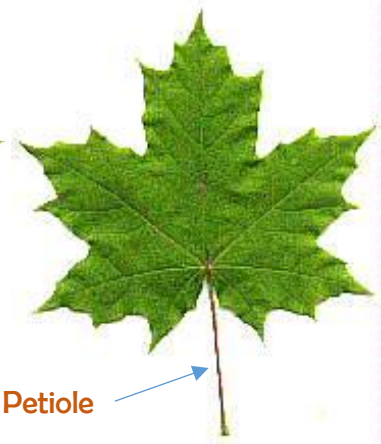


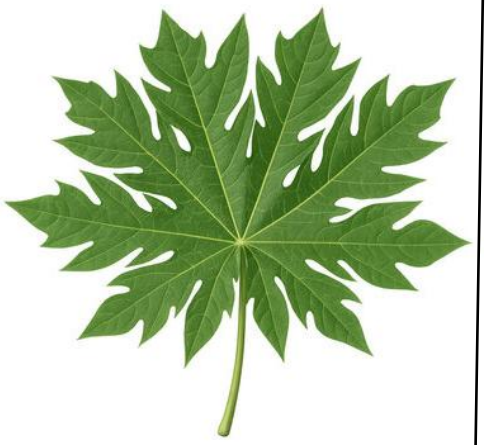
Simple



Pinnately Lobed



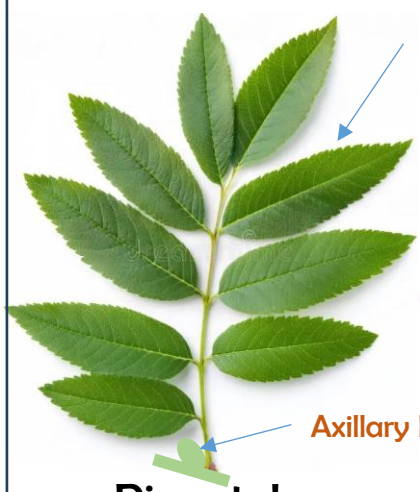




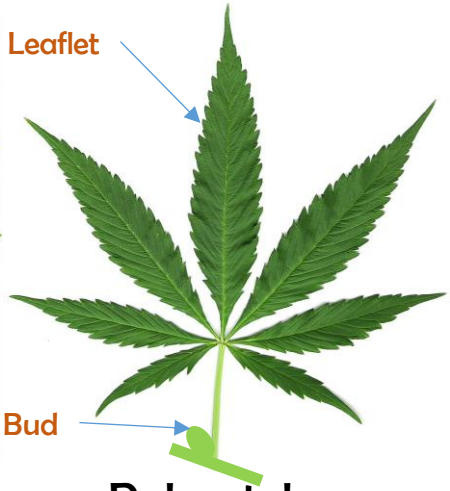
Palmately Lobed



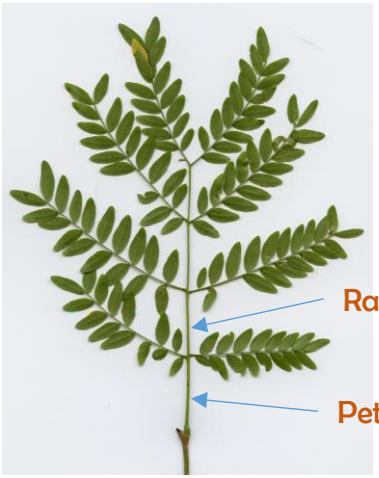
Doubly Pinnately Lobed



Pinnately Compound



Palmately Compound



Doubly Pinnately Compound

What Leaf shapes, divisions, arrangements, attachments and margins do you see?

A



C





B



D

Flowers

- The sexually reproducing part of the plant

- A flower is a modified shoot system bearing one or more stamens and/or carpels, usually both, and a perianth of petals and/or sepals, all of which are highly modified leaves



Pholistoma auritum – Fiestaflower

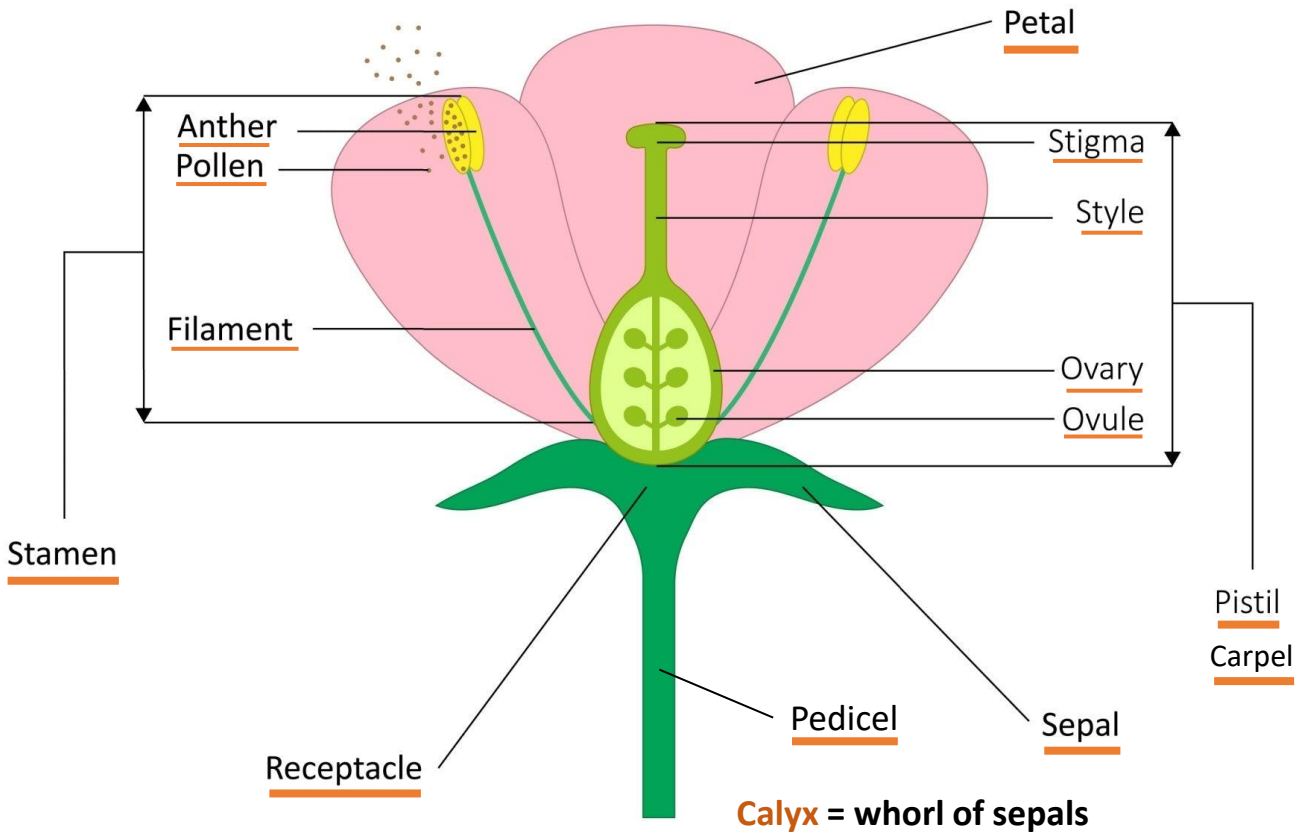
Model Flower

- Pistil/Carpel (f
  - Ovary (with o
  - Style
  - Stigma

Gynoecium = who
stamens

- Stamen (male
  - Filament
  - Anthers
  - Pollen

Androecium = wh
pistils



Calyx = whorl of sepals
Corolla = whorl of petals
Perianth = calyx + corolla

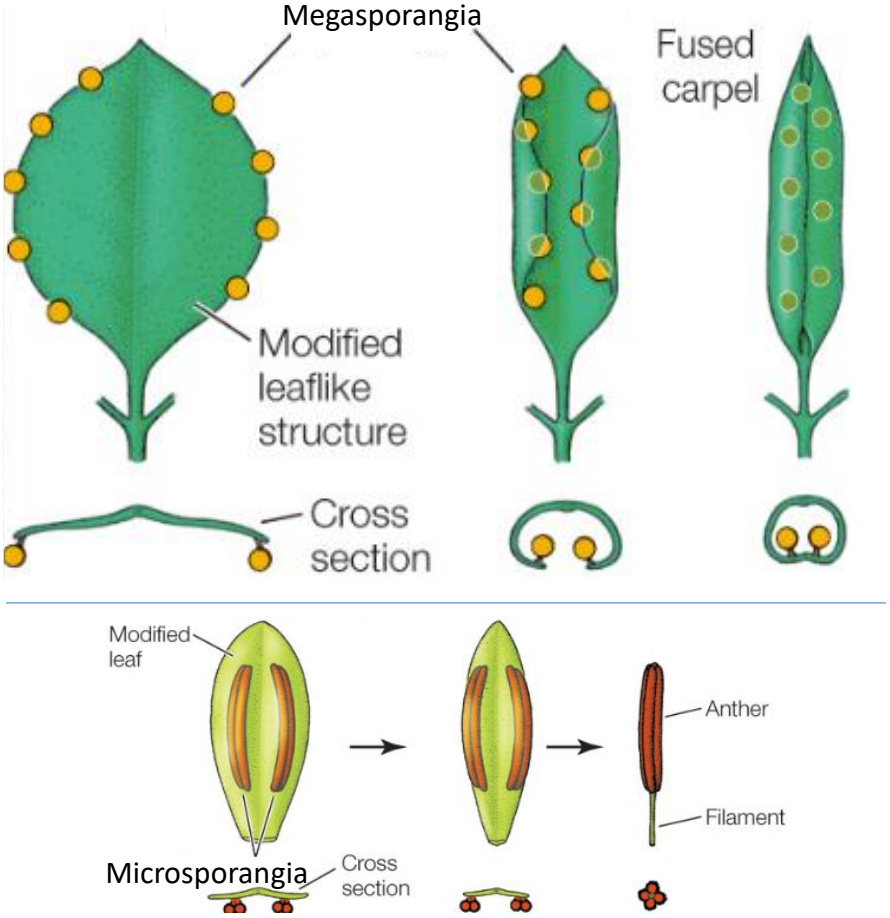
Do You See The Distinct Parts?



Ivesia arizonica

Carpel Evolution

- The carpel is what defines a flowering plant, or angiosperm
- About 140 million years ago, the first flowering plants emerged with fusion of the leaf margins, encasing *and protecting* their developing ovules and, following fertilization, the seeds
- Stamens and other flower parts are also modified leaves



Important Processes in Evolution:
Fusion

The union of plant parts that were originally separate

- Connate: Fusion of like parts - i.e. fusion of all the petals

- Adnate: Fusion of unlike parts - i.e. stamens to petals

Datura wrightii



Penstemon petiolatus



Stamens fused with petals = epipetalous

Fusion: There are many (adjacent) things that can be fused

The hypanthium is the fusion of sepal, petals, stamens, and carpel



Agave sp.

The pistil (gynoecium) can be a complex of two or more carpels fused together





Calochortus nuttalii

Back to Important Processes in Evolution: Reduction

Parts of the flower have become smaller, or completely missing

-> What is "missing" from these flowers?



Boerhavia gracillima



Populus freemontii

Important Processes in Evolution:
Elaboration

Parts of the flower have become more complex, specialized, detailed

spur (nectary)



petals

Delphinium parishii

bearded filaments



Tradescantia occidentalis



corona

Passiflora arizonica

Knowledge Check

Penstemon:
Fusion? Reduction? Elaboration?



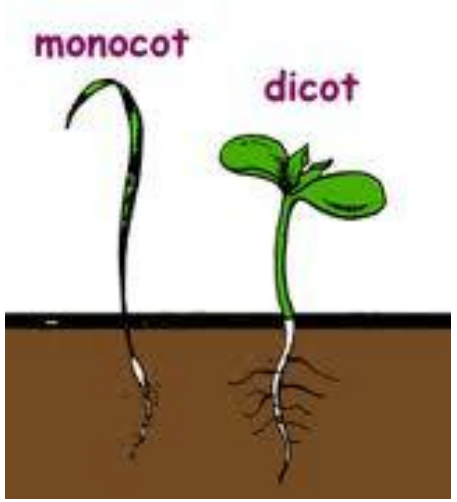
Penstemon barbatus



Euphorbia spathulata

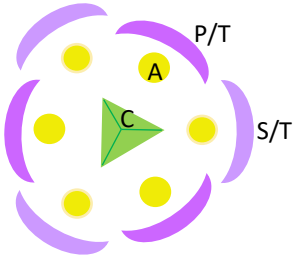
Monocot and (Eu)dicots

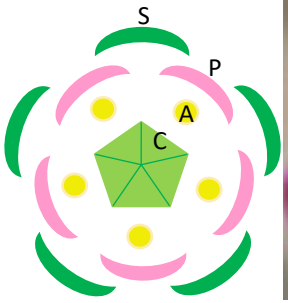
- Monocot– a plant with one seed leaf (cotyledon). Parallel leaf veins, Flower parts in 3's. Lily, Agave, Grass.
- Dicot– a plant with two cotyledons. Net-like leaf veins, Flower parts in 5's (and sometimes 2's or 4's). Rose, Mustard, Bean, Hibiscus.





Monocot flower







Dicot flower

Flower Symmetry

Radial Symmetry: Flower has same size petals and parts; it can be cut like a pie into identical equal pieces

i.e., Rose



Regular Flower: *Rosa woodsii*

Bilateral Symmetry: Flower has two or more different shapes, sizes, colors but can be divided by a single plane into two mirror-image halves like a human face

i.e., Orchid, Lupine



Irregular Flower: *Lupinus sparsiflorus*

Complete and Incomplete Flowers

Potentilla sp.



Complete flowers have all of the following:

- Sepals
- Petals
- Stamens
- Pistil

Incomplete flowers are missing one of these

Aristolochia watsonii



Floral tube of fused sepals. No petals.

-> Incomplete flowers are examples of what?

Perfect and Imperfect Flowers

Perfect flowers are bisexual

- Stamens and Pistils



Oenothera caespitosum
(Evening Primrose)

Imperfect flowers

- Male or female

staminate



pistillate



Simmondsia chinensis
(Jojoba)

-> Incomplete flowers are examples of what evolutionary process?

One, Two, and Same "Houses"

Monoecious – Plants have both male (staminate) and female (pistillate) flowers on the same plant



Ambrosia deltoidea

Dioecious – Plants have staminate and pistillate flowers on separate plants



Bursera microphylla (staminate)

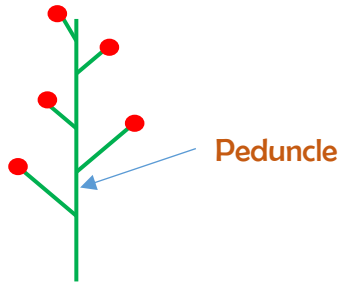


Bursera microphylla (pistillate)

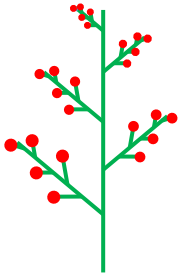
Synecious – Plants have perfect flowers (bisexual)

Inflorescence

The arrangement or cluster of flowers on a single main stem (peduncle) or a complex branch system



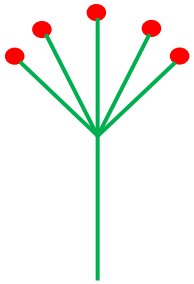
Raceme



Panicle

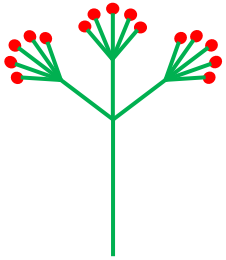


Spike

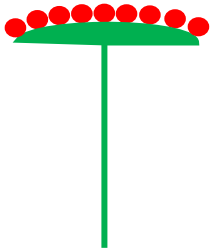


Umbel

Indeterminate

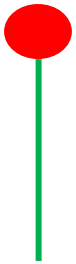


Compound Umbel

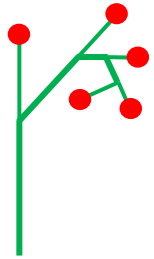


Capitulum

Determinate



Solitary

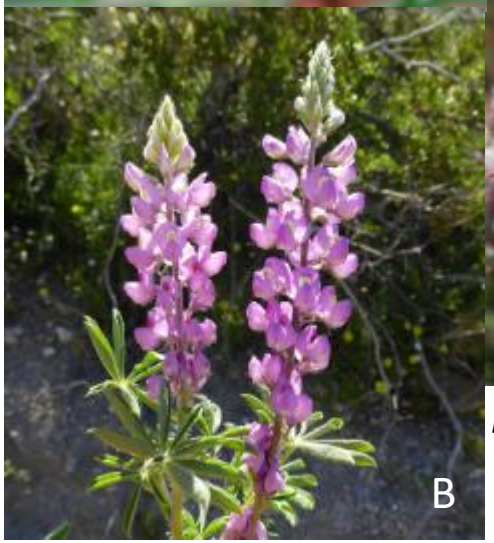


Cyme
(Scorpioid)

What inflorescence type do you see?

Lepidium lasiocarpum





Asclepias subulata





Eriogonum fasciculatum

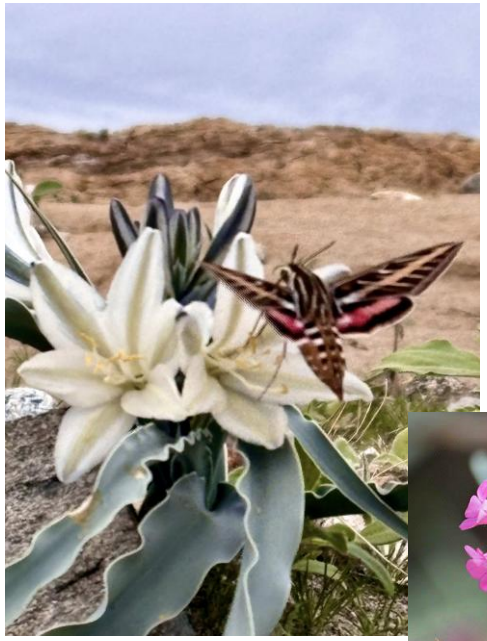
Lupinus arizonicus



Solidago sp.

Flowers and Pollination

The color, shape, size and movement of flowers and their parts often reflect their mode of pollination



Hesperocallis undulata





Penstemon parryi



Machaeranthera tanacetifolia

Fruit

- A fruit is a ripened ovary
- It protects seeds and aids in dispersal

capsule
(*Eschscholzia sp.*)



cypsela
(*Uropappus lindleyi*)



capsule
(*Proboscidea sp.*)



legume
(*Parkinsonia microphylla*)



bur/nut
(*Krameria bicolor*)



berry
(*Opuntia sp.*)



Adaptations of Desert Plants



Microphylly
Little leaves



Succulence
Water storage tissue



Pubescence
Reflective/insulating hairs



Stem Photosynthesis



Drought Avoidance
Annuals

Epicuticular wax

CAM Photosynthesis

Drought/heat
Deciduousness

Etc.

Plant Phenology in Northern Sonoran Desert

Phenology in the desert is governed by temperature, photoperiod and precipitation. Precipitation here occurs over two seasons.

- Winter-Spring (Dec-Apr) – Driven by wide-spread cool weather systems
  Summer-Fall (Jul-Sep) – Driven by localized, warm monsoon storms
- Both wet seasons have a cohort of annuals that germinate and perennials that leaf out
- Some plants respond to particular timing and duration rains
- Still others, like cactus and other succulents have fairly consistent phenology regardless
- Rapidly changing climate is shifting/changing the seasons

Oenothera primiveris *Gilia flavocincta* *Phacelia distans*



Allionia incarnata *Kalstroemia grandiflora*



Proboscidea parviflora



Lupinus sparsiflorous



Eschscholzia californica subsp. mexicana

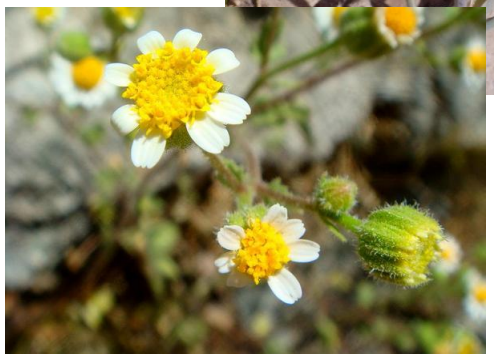
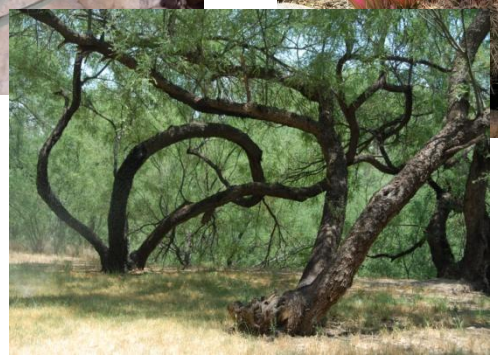
A Well Deserved Break Now!



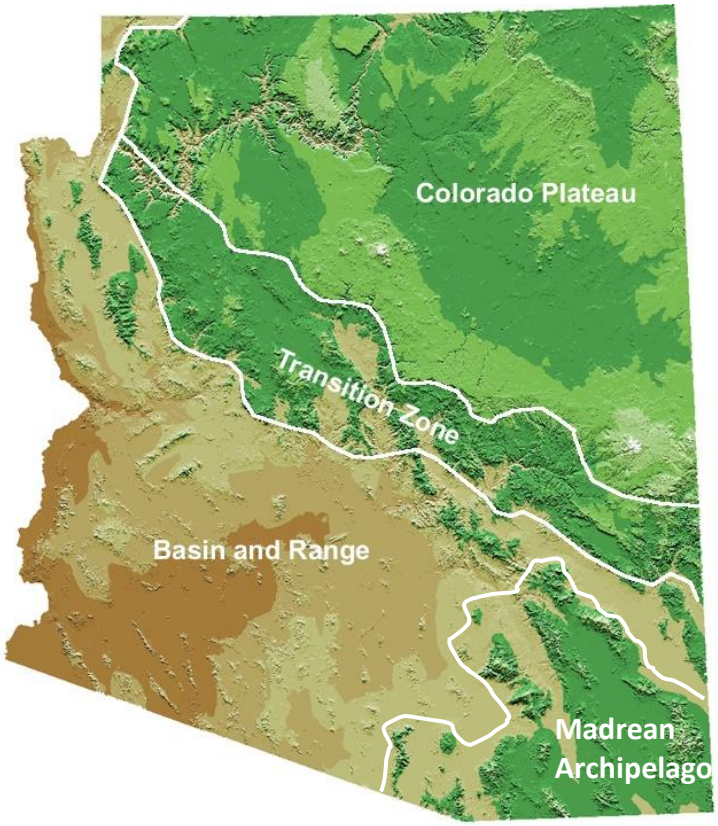
Arizona and biodiversity

Arizona is a region of great plant diversity - ca 4000+ species of vascular plants, 4th in total number of plant species in the country, behind California, Texas and Florida




Geologic Provinces of Arizona



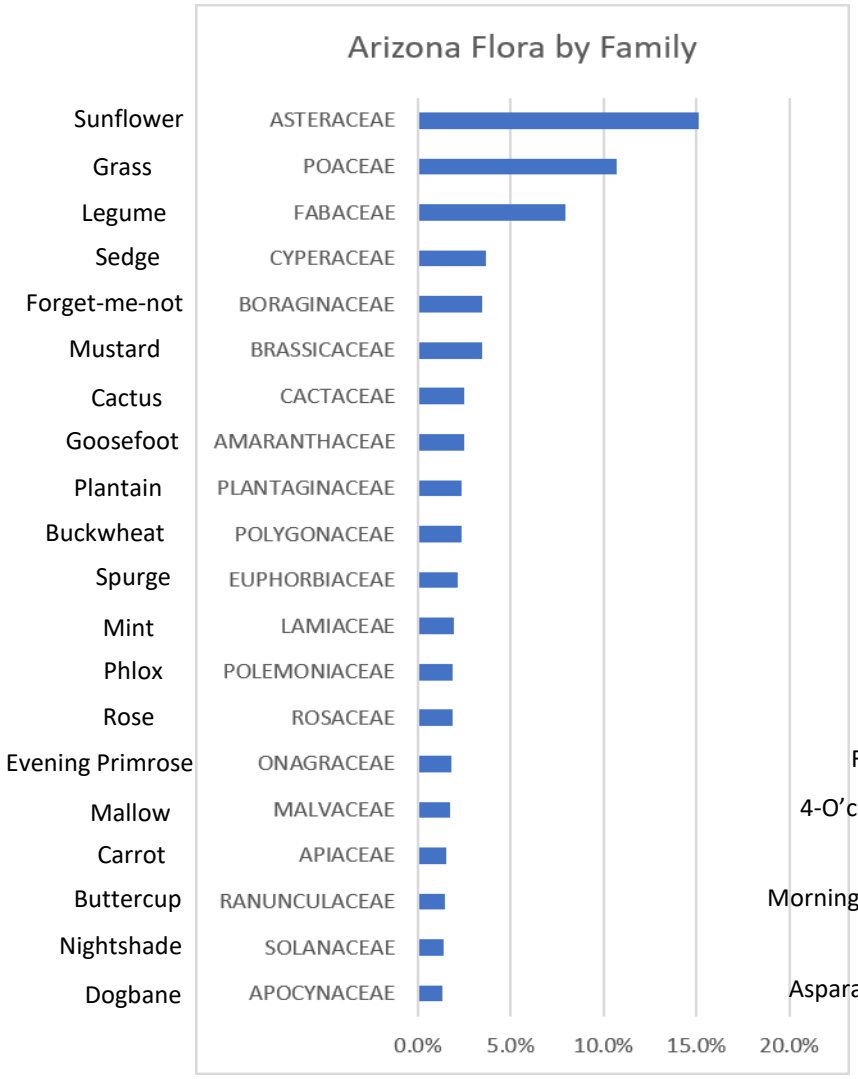
Colorado Plateau: A mile high block of earth crust forming an arid, cold desert with higher elevations of pinion-juniper woodlands and peaks with mixed conifers.

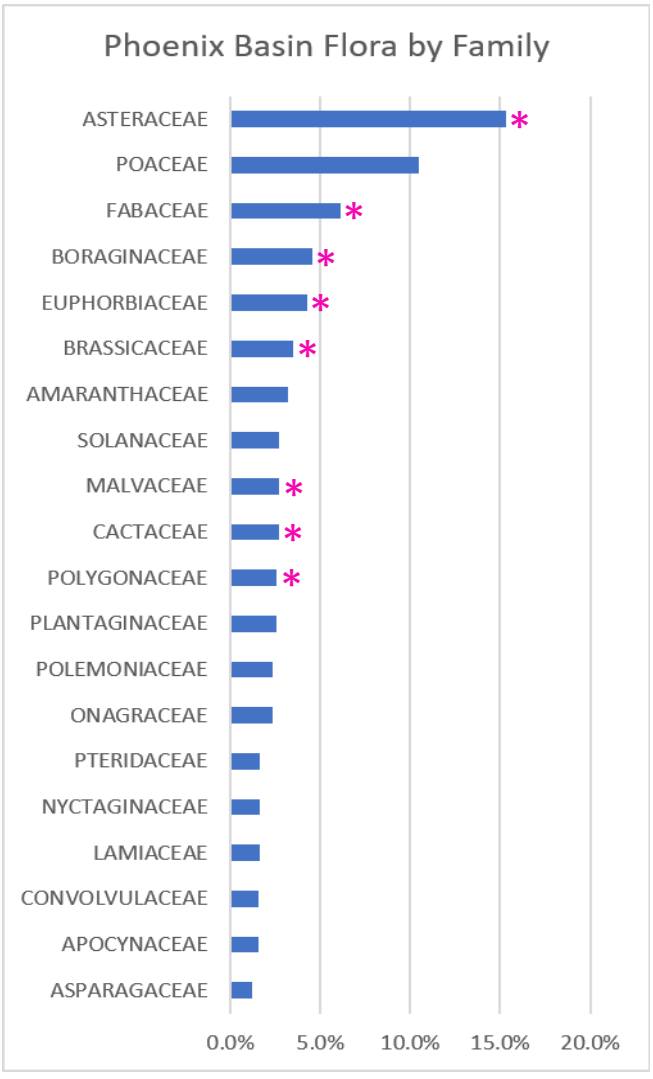
Basin and Range: Region of tectonic extension featuring valleys separated by sharp, mostly low, mountain ranges with northern Sonoran Desert flora transitioning to Mohave Desert flora in the northwest.

Transition Zone: A mountainous strip between the Colorado Plateau and the Basin and Range featuring a mix of oak and juniper woodlands, interior chaparral, and semi-desert grasslands. A region of high endemism.

Madrean Archipelago: Another region of tectonic extension. A biologically diverse area due to the convergence of the Sonoran and Chihuahuan deserts with the Rocky Mountains and Sierra Madre giving rise to high endemism.

Arizona's Top Plant Families





Plantain – Plantaginaceae



P. fendleri

Parry's Penstemon

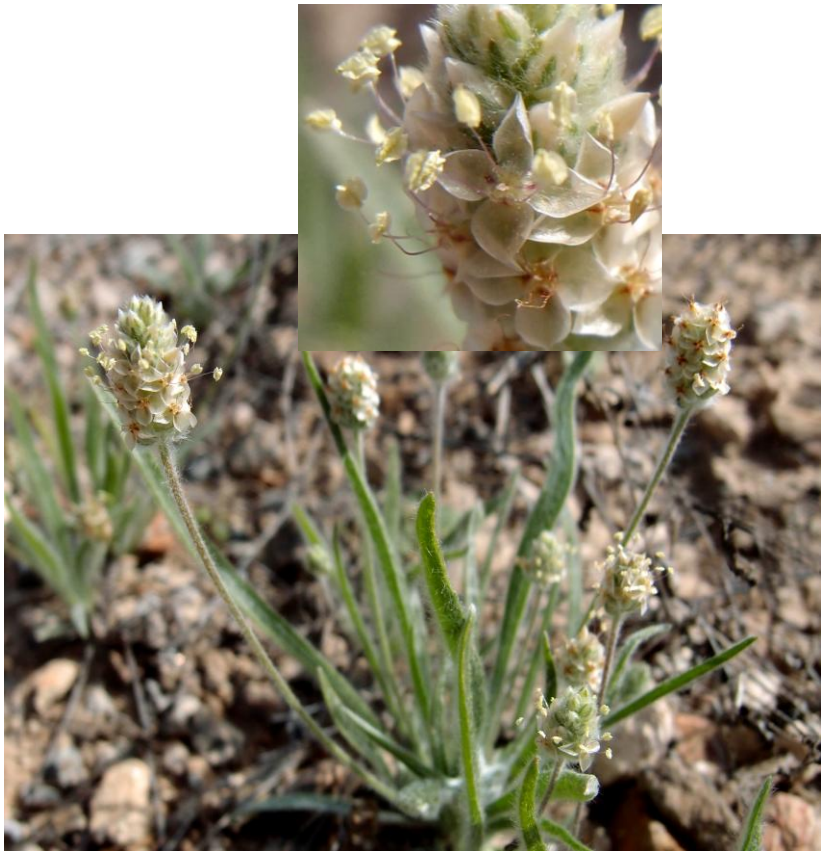


Eaton's Penstemon

- Plants mostly herbaceous, a few are shrubs
- Bilaterally symmetric, tubular flowers of 5 fused petals appearing as lobes with 2 upper and 3 lower
- 4 stamens, two ranked, sometimes a 5th, sterile (staminode)
- Inflorescence a racemose spike

Plantain

- Simple leaves, usually a basal rosette



Wooly Plantain

Monkey Flower



Violet Snapdragon





Twining Snapdragon

Mint - Lamiaceae



Chia







Desert Lavender

- Bilabiate flowers of fused petals, (2 upper and 3 lower), with the lower lobe often elongated and elaborate. Flowers of most species are small
- 5 fused sepals separating at the tips into points
- 4 stamens, 2 ranked
- Dry fruit separating into up to 4 nutlets
- Inflorescence of congested cymes at the nodes

Mint

- Leaves simple, in opposite decussate pairs, deltoid, and serrate
- Stems often have a square cross section
- Leaves often fragrant when crushed
- Rolling the stem in your fingers is a good test



Texas Betony



Mexican Skullcap

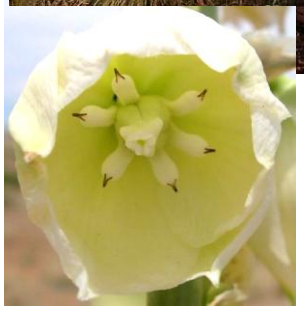
Asparagus - Asparagaceae





Bluedicks





Soaptree Yucca





Golden Agave

- Perennial, monocot herbs or shrubs
- Radially symmetric flowers with 6 tepals in 2 whorls of 3
- 6 stamens in 2 whorls matching tepals, adnate
- 3-locular ovary with 3-lobed stigma
- Leaves are a basal, often a rosette, linear
- Vegetative reproduction by rhizomatous offsets

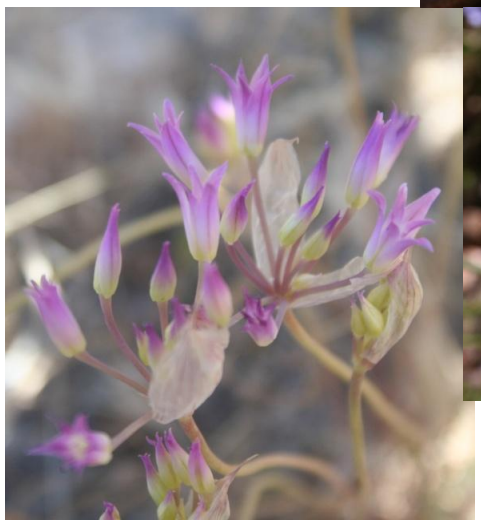
Amaryllidaceae

- Herbaceous perennial, monocot geophytes. All feature an underground stem in the form of a bulb or corm
- Regular flowers with 6 tepals in 2 ranks of 3
- 6 stamens arranged within corolla in 2 ranks of 3
- 3-locular ovary with 3-lobed stigma
- Inflorescence an umbel subtended by a spathe
- Leaves are a basal, linear, grass-like



Desert Onion





Tapertip Onion



False Garlic

Poppy - Papaveraceae



Mexican Poppy



Prickly Poppy



Cream Cups

- Herbaceous annuals, mostly, often with milky sap
- Flowers often solitary, showy with 4 to 6+ petals
- 2 or 3 sepals sometimes fused and quickly deciduous
- Many stamens
- 2 to many fused carpels
- Fruit a dry capsule dehiscing longitudinally or poricidally under persistent stigmatic disk
- Leaves pinnately lobed to heavily dissected

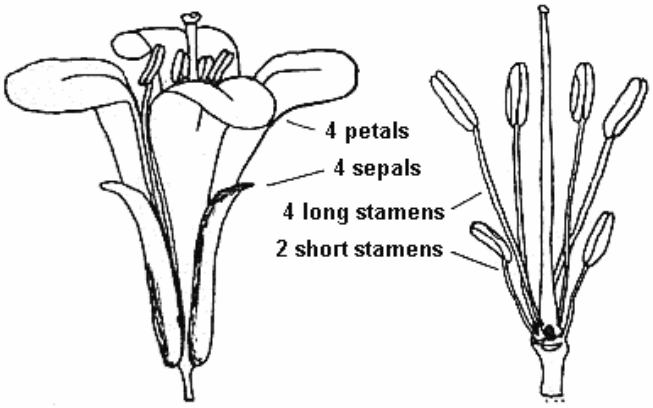
Mustard - Brassicaceae

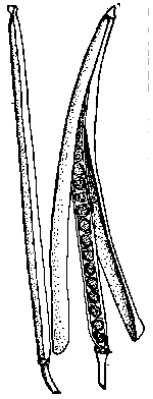


Wallflower



Peppergrass





Silique

- Herbaceous annuals and perennials
- Flowers with 4 petals (Formerly called Cruciferae – cross)
- Inflorescence a raceme
- Four unfused (distinct) sepals
- Six stamens, with 4 taller and 2 shorter
- Leaves alternate, simple to pinnately lobed
- Fruit is a silique

Mustard

- A large family with over 100 species in Arizona. Desert members are usually yellow or white
- Several old-world species such as Sahara Mustard, once introduced to a landscape soon dominate



Yellow Bladderpod

Purple Bladderpod

Perennial Rock Cress

Forget-me-not - Boraginaceae



Fiddleneck





Cryptantha

Combseed



- Usually annuals in the desert
- Small, 5-lobed funnel-formed flowers, often featuring contrasting color appendages
- Color usually yellow or white in deserts
- Inflorescence is a coiled cyme bearing flowers on the upper side, maturing as it uncoils
- Plants and fruit bristly

Forget-me-not

Wild Heliotrope



Scorpion Weed

- Plants of this family comprise the majority of spring desert annual groundcover

Mallow- Malvaceae

Globe Mallow





Abutilon



Coulter's Hibiscus



Velvet Mallow

- Often perennial shrubs and subshrubs with a few herbaceous annuals
- Flowers with 5 separate petals. Usually showy
- Many stamens fused together as a column surrounding the pistil(s), separating below. One to many stigmas separating above
- Five partially fused sepals. Often with one or more small bracts underneath (subtending)

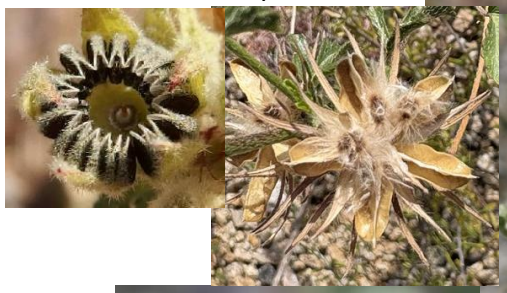
Mallow

- Pistil of several fused carpels, the styles separating above the stamens
- Fruit a schizocarp
- Leaves are usually deltoid, simple to lobed with dentate or serrate margins and often covered with stellate hairs
- Fruit is a schizocarp
- 50+ members in Arizona
- This family of plants includes cotton, marshmallow, and okra

Desert Rose Mallow



Schizocarps





Desert Hibiscus

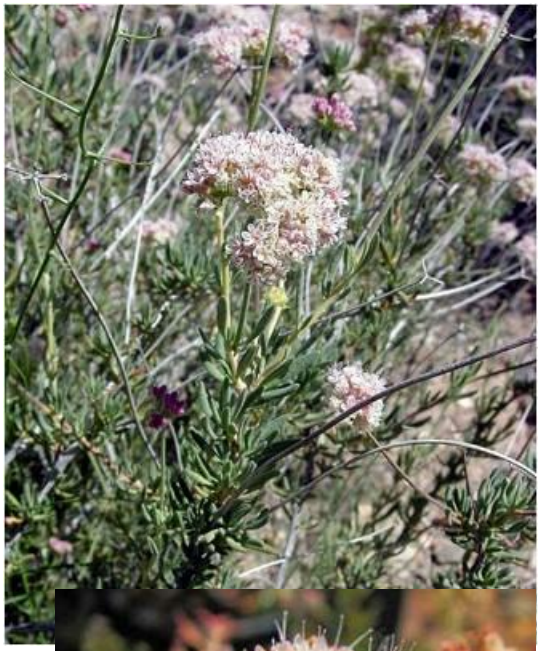


Buckwheat - Polygonaceae





Desert Trumpet





Flat Top Buckwheat

- Plants can be herbaceous annuals to woody perennials
- Flowers of 3-6 tepals, yellow to white to pink
- Stamens usually as many or more than tepals
- One pistil of two or more fused carpels with one to several styles
- Inflorescence racemose to umbelliform
- Leaves simple

Buckwheat





Chorizanthe



Desert Rhubarb



Skeleton Weed

- Non-native Buckwheat used to be a common poor soil adapted crop plant that went out of favor with nitrogen fertilizers

4 O'Clock - Nyctaginaceae

Desert Wishbone Bush



Angel Trumpet





Trailing Windmills

- Commonly both annuals and perennials.
- Sepalous flowers of 5 or more sepals, usually forming a funnel, subtended by a calyx-like involucre of bracts.
- The lower part of the flower often hardens to form a protective shell for the fruits.

4 O'Clock



Coulter's Spiderling







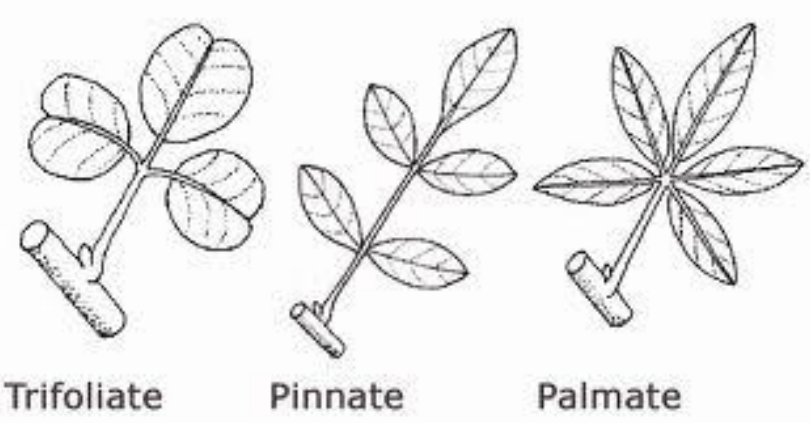
Scarlet Spiderling

- Many Perennials are biseasonal bloomers while most annuals are summer annuals.
- 4 O'clocks are a favorite food of the Sphinx Moth larvae whose populations explode during wet summer seasons when the plants are abundant.

Peas - Fabaceae

- Herbs, shrubs and trees
- Flowers of 5 petals, but variable form
- 10 stamens, generally
- One carpel. Fruit is a 2-valved pod
- Leaves compound
- Nitrogen fixing root nodules host rhizobia that manufacture NH_3
- Very large family divided into 3 (6) subfamilies: Caesalpinioideae, Mimosoideae, and Papilionoideae

Fabaceae feature compound leaves




Papilionoid


Caesalpinioid

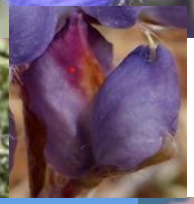

Mimosoid

Papilionoideae



Lupine

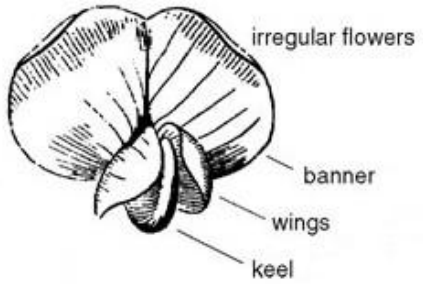




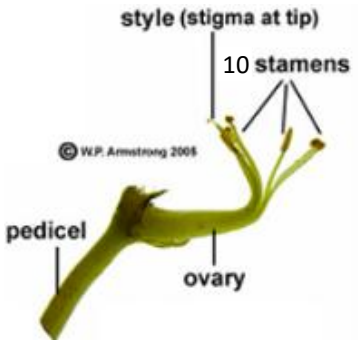


Ironwood





- Flowers irregular, upper petal is large (banner) lower two petals fused into a keel
- Leaves pinnate, trifoliate or palmately compound
- Stamens fused about the pistil. Upturned within the keel
- Plants common in this subfamily: Lupine, Ironwood, Astragalus, Peas



Caesalpinioideae

Palo Verde



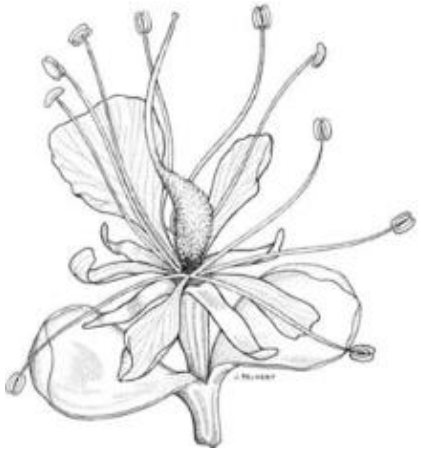






Desert Senna

- Usually woody shrubs, trees
- 5-merous flowers slightly irregular, petals distinct, uppermost usually smaller
- Fusion in lower calyx
- Inflorescence is often a raceme or panicle of flowers
- Leaves usually once or twice pinnately compound



Mimosoideae

Acacia

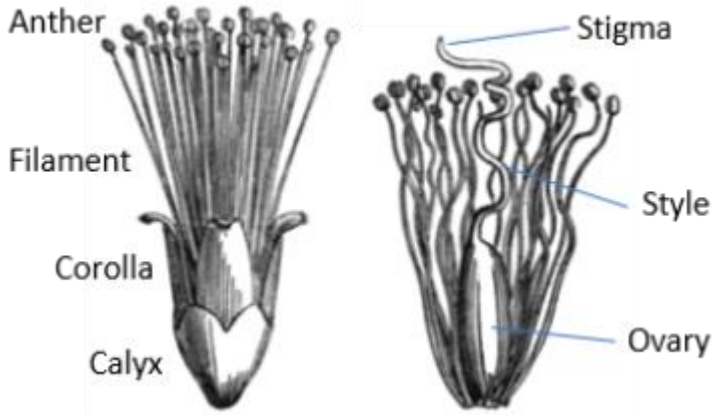






Fairy Duster

- Leaves usually twice pinnately compound
- Flowers regular, petals small, inconspicuous. Partially fused calyx and corolla
- 10 (or more) highly exserted, colorful stamens
- Inflorescence is a head or spike-like cluster of flowers



Sunflower - Asteraceae

Tansyleaf Aster





- A dense, composite inflorescence of many small flowers resembling a single flower
- The disk (center) consists of many small, fertile flowers
- The rays ("petals") are showy flowers. In some species pistillate, in others infertile
- Leaves alternate or opposite, lobed or dissected

Desert Marigold

Sunflower Anatomy



Ray Flowers

Disk Flowers





Receptacle

2 Lobed Style

5 Fused Stamens

5 Lobed Corolla

Pappus

Ovary



Disk Flower

Phyllaries

Vestigial Pistil



Ray Flower

Sunflower

- Some species have only disk flowers
- Some species consist of ligulate flowers: fertile, bisexual ray-like flowers



New Mexican Thistle



Poreleaf



Ligulate Flower

Secondary pollen presentation
"plunger mechanism"



Desert Chicory



Wire-Lettuce

Sunflower

- Some sunflowers are wind pollinated and do not feature showy flowers
  - These are usually monoecious or diecious



White Sagebrush



Triangle Bursage

Spurge - Euphorbiaceae

- A special inflorescence called a cyathium with staminate flower (one or more), and one pistillate flower
- The staminate flower(s) are reduced to a single stamen
- The pistillate flowers are reduced to a fused, 3-carpellate ovary
- No petals or sepals, but some with subtending bracts (cyathophylls)





Whitemargin Spurge

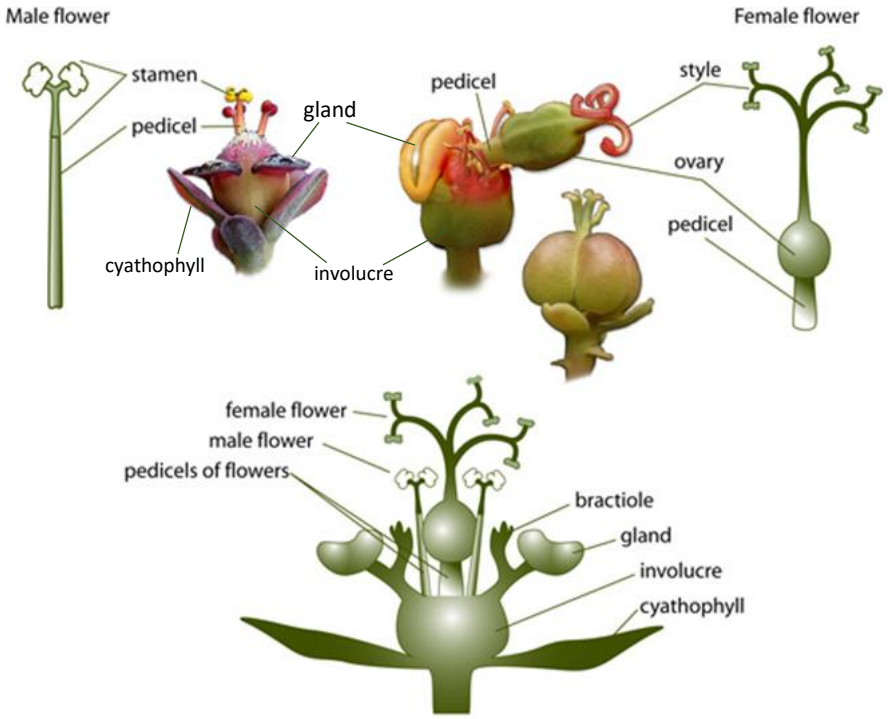




Ditaxis neomexicana

Spurge





- One or more nectary glands attached to the cyathium
- The glands in some genera may have a white appendage extending radially (Chamaesyce)
- All species ooze milky, latex sap if cut
- Fruits are capsules of seeds (three) that explode when ripe
- Some old-world plants are succulent and are confused with cacti (convergent evolution)

Cactus - Cactaceae

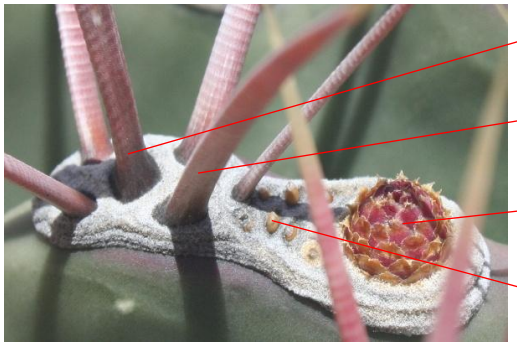




Cylindropuntia bigelovii
Carnegiea gigantea
Opuntia engelmannii
Ferocactus cylindraceus







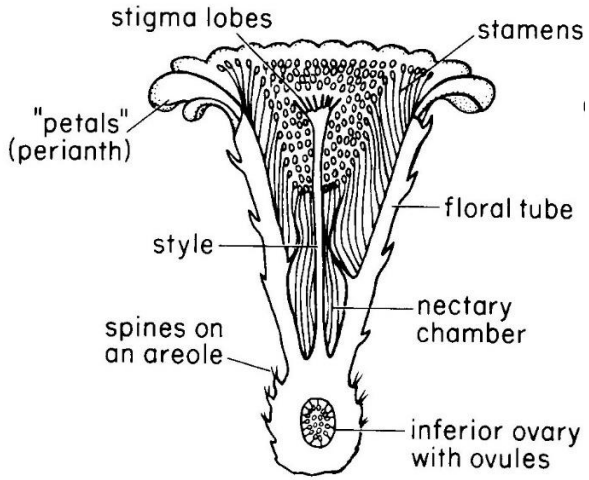
Areole

- Succulents. Photosynthetic stems forming small to large shrubs and "trees"
- The areole is a highly modified and condensed branch
- From areoles emerge other branches (arms), leaves (spines) and flowers
- Cactus spines are modified leaves

Cactus - Flower



Hedgehog Cactus



- Flowers of many tepals. Outer tepals often sepalous. Inner tepals petalous
- A specialized outer layer of fused receptacle and stem tissue enclose the ovary (pericarpal) which elongates to house much of the flower
- Many stamens born on the floral tube
- Ovary of few to many carpels
- One stigma, often many lobed
- Spines usually persistent or areole deciduous

Cactus - Columnar

Barrel Cactus



Hedgehog Cactus



Saguaro



Pincushion Cactus



Cactus - Jointed

Buckhorn Cholla



Teddy Bear Cholla



Christmas Cholla



Prickly Pear



Honeysuckle - Acanthaceae

- Perennials, herbaceous or shrubs woody at the base
- Inflorescence variable but often racemose
- Flowers usually tubular, bilabiate, bilaterally symmetric, white to red
- 4 stamens, 2-ranked
- Fruit an exploding capsule with 2 to 4 flat seeds

• Most are showy hummingbird attractors



Desert Honeysuckle



Chuperosa





Arizona Carlowright

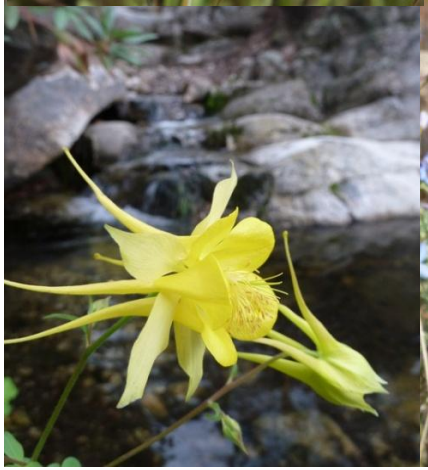
Buttercup - Ranunculaceae

Desert Anemone









Yellow Columbine

Parish Larkspur

- Mostly herbaceous “root” perennials, but a few annuals
- 5 petaloid sepals and 0 to many petals
- Many stamens, spirally arranged
- Multiple pistils (3 to many)
- Leaves usually palmately lobed to deeply dissected
- Spurs of Columbine and Larkspur are nectaries
- Anemones offer pollen reward only

Phlox - Polemoniaceae

Lesser Yellowthroat Gilia



Miniature Woolstar







Desert Phlox

- Herbaceous annuals and perennials
- Inflorescence a compound cyme – appears paniculate
- 5 petals and sepals both fused from the base, each opening into 5 lobes
- 5 stamens fused to the corolla tube
- 3-loculed ovary, one style, 3-lobed stigma
- Leaves simple to pinnately lobed, leaves and lobes narrow

Morning Glory - Convolvulaceae



Scarlet Creeper





Dodder

Arizona Blue Eyes



- Herbaceous shrubs and vines
- Flowers funnel-formed, 5 lobed, often very showy. White, violet, to red in native plants
- Stems herbaceous often viney
- Leaves cordate/sagittate to deeply lobed in the viny species, linear to absent in some of the low growing plants

Nightshade - Solanaceae

Silver Nightshade





Sacred Datura

- Mostly herbaceous plants, sometimes a woody shrub (Wolfberry)
- Corolla of 5 petals fused, funnel-form. Calyx similarly fused, but persistent
- Inflorescence a scorpioid cyme, often reduced to solitary in axils
- 5 stamens, adnate to corolla
- Pistil of 2 fused carpels. Fruit a fleshy berry or dry capsule

Nightshade

Berlander's Wolfberry







Desert Tobacco

- Leaves simple, alternate, fascicled in *Lcyium*
- Many members are poisonous to varying degrees. Tobacco mildly, Datura significantly
- Several plants of this family are important agricultural crops (i.e. tomato, potato, chiles)

Caltrop – Zygophyllaceae



Creosote



Arizona Poppies



Puncture Vine



Fagonia

- Shrubs and herbaceous annuals and perennials
- Flowers usually with 5 distinct petals/sepals. Solitary in the axils
- Stamens usually twice as many as petals in 2 whorls
- A 5 to 10 loculated carpel - schizocarp
- Compound leaves, usually opposite and pinnate